

Note on gpt.py computational workflow.

Computational Modules.

- RMS norm: normalization of token's embedding vector $\mathbf{x} \in \mathbb{R}^d$. There are no learnable params in RMSNorm. This is different from transformer paper's LayerNorm.

$$\text{RMSNorm}(\mathbf{x}) = \frac{\mathbf{x}}{\text{RMS}(\mathbf{x}) + \varepsilon}, \quad \text{RMS}(\mathbf{x}) = \sqrt{\sum_{i=1}^d x_i^2}, \quad \mathbf{x} = (x_1, \dots, x_d), \quad \varepsilon > 0.$$

- Rotary Positional Embedding (RoPE): positional embedding using sin, cos positional tensors. This is different from transformer paper's Positional Embedding.

Input x (which is the Q, K in causal attention module) with shape (B, T, H, D) where B = batch size, T = sequence length, H = number of heads in multi-head attention, D = dimension of each head (must be even). Let $\sin = (\sin^{t,i})_{0 \leq t \leq T-1, 0 \leq i \leq D/2}$, $\cos = (\cos^{t,i})_{0 \leq t \leq T-1, 0 \leq i \leq D/2}$ be tensors each with shape $(1, T, 1, D/2)$. Then for each batch b , each index $t \in \{0, \dots, T-1\}$, each head h , RoPE does the following:

$$(x^{b,t,h,0}, \dots, x^{b,t,h,D-1}) \rightarrow \begin{cases} \cos^{t,i} x^{b,t,h,i} + \sin^{t,i} x^{b,t,h,i+D/2} & \text{for } 0 \leq i \leq D/2 - 1, \\ -\sin^{t,i-D/2} x^{b,t,h,i-D/2} + \cos^{t,i-D/2} x^{b,t,h,i} & \text{for } D/2 \leq i \leq D-1. \end{cases}$$

Position information, originally in t , is encoded in such rotation via $\cos, \sin$. Usually we take $\cos^{t,i} = \cos\left(\frac{t}{10000^{2i/D}}\right)$ and $\sin^{t,i} = \sin\left(\frac{t}{10000^{2i/D}}\right)$.

- Causal Self-Attention. The self-attention module with specific causal mask construction.

- number of heads `n_head % number of kv heads n_kv_head = 0`.

- self-attention. Input x has shape $(B, T, _)$, which stands for the current forward pass. Then Q, K, V are from input x , i.e., $Q = W_Q^T X$ etc. They are re-shaped into shape $(B, T, \text{n_head}, \text{head_dim})$ for Q and $(B, T, \text{n_kv_head}, \text{head_dim})$ for K, V , respectively. Then apply RoPE to Q, K , apply RMSNorm to Q, K , finally apply $(1, 2)$ transpose $(B, T, H, D) \rightarrow (B, H, T, D)$ to Q, K, V , i.e., to process each attention head to the full batch (head-wise parallelization);

- kv-cache and multi-query-attention. T in (B, H, T, D) stands for the number of queries. For Q , use number of queries in this forward pass = T_q ; for K, V , after kv-cache, use number of queries in the cache + current forward pass = $T_q + (T_k)_{\text{prev}} = T_k$.

PyTorch 2.0's `scaled_product_attention`:

```
def scaled_dot_product_attention(
    query: Tensor,
```

```

key: Tensor,
value: Tensor,
attn_mask: Tensor | None = None,
dropout_p: float = 0.0,
is_causal: bool = False,
scale: float | None = None,
enable_gqa: bool = False,
) -> Tensor: ...

```

Multi-Query-Attention (Group-Query-Attention, gqa):

`enable_gqa = self.n_head != self.n_kv_head`: multiple query heads use the same key/value head, or in other words duplicately use key/value head for a chunk of query heads. This saves computation. Example: `n_head=16`, `n_kv_head=4`, then each group of 4 query heads share the same kv head, or the same kv head repeatedly used 4 times for each chunk of 4 query heads.

Mask construction in multi-query-attention: Discuss the following cases

- (1) Inference with no increment, i.e. infer the whole sequence at once ($T_q = T_k$) or training (kv cache is empty): Auto regressive case. Each query only attends keys/values before it. Use standard causal mask. See Figure 1 left.
- (2) $T_q = 1$, inference with single query in this forward pass. Each query attends all the keys/values in the cache. No mask needed. See Figure 1 middle.
- (3) Not (1) or (2): $1 < T_q < T_k$. Inference with a chunk of T_q queries in this forward pass. Each query attends all the keys/values in the old kv cache $(T_k)_{\text{prev}} = T_k - T_q$, but for the newly added T_q items of kv, i.e. within the chunk we use causal attention so that each query only attends keys/values before it. See Figure 1 right.

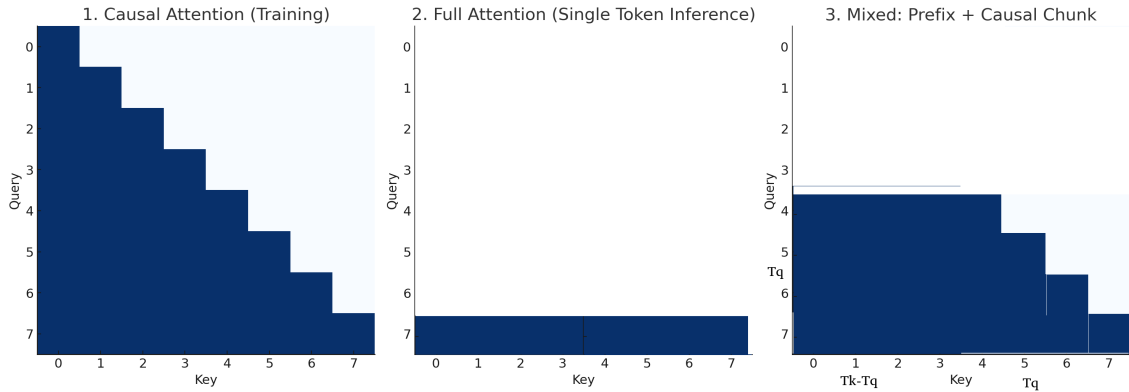


Figure 1: MQA masks.

- final projection. Concatenate heads

$$(B, n_head, T, head_dim) \rightarrow (B, T, n_head, head_dim) \rightarrow (B, T, n_head \times head_dim) .$$

Followed by a linear layer $n_head \times head_dim \rightarrow n_embd$.

- MLP. Layer structure: $n_embd \rightarrow 4 \times n_embd \xrightarrow{\text{ReLU}^2} n_embd$. Activation function use

$$\text{ReLU}^2(x) = \begin{cases} x^2 & \text{if } x \geq 0 , \\ 0 & \text{if } x < 0 . \end{cases}$$

- Block. Use residual connections

$$x \mapsto x' = x + \text{Causal Self-Attention} \mapsto x'' = x' + \text{MLP} .$$

The whole GPT workflow.

- Network architecture:
 - **wte**: word-to-embedding layer.
 - **h**: transformer block layer.
 - **lm_head**: language modeling head), head layer, embedding $\rightarrow$ token logits.

Input ids

↓

Embedding (wte)

↓

Norm

↓

$N \times$ Transformer Blocks (Block) (N=12)

↓

Norm

↓

Linear Output Head (lm_head)

↓

Logits / Loss

- Initialize weights:
 - linear layers: $\mathcal{N}(0, \sigma)$ where $\sigma = \frac{1}{\sqrt{fan_in}} \cdot \left(1 \wedge \sqrt{\frac{fan_out}{fan_in}}\right)$
 - embedding layers: $\mathcal{N}(0, 1)$
 - **lm_head** weights: 0
 - **c_proj** weights:
 - initialize the rotary embeddings:
 - Optimizers: AdamW for embedding layers and Muon for linear layers.
 - Naive autoregressive streaming inference: generate the next token through output logits.